GenoFold: A Trained, Confidence-Aware, and Transferable Framework for 3D Genome Architecture Reconstruction

Anonymous Author(s)
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Author names and affiliations withheld

***Abstract*—The three-dimensional organization of the genome governs enhancer–promoter contact, the placement of topologically associating domain (TAD) boundaries, and the partition of the nucleus into active and inactive compartments. Computational reconstruction of this architecture from Hi-C data has advanced over the past decade, yet three limitations remain common: methods take a single input modality, they emit coordinates with no calibrated measure of reliability, and their preprocessing, inference, scoring, and visualization stages live in separate tools. We present GenoFold, a framework that integrates Hi-C contact matrices, per-bin sequencing coverage, sequence composition, and genomic metadata as co-equal inputs and reconstructs chromatin structure through a single six-stage pipeline. GenoFold encodes genomic loci as a contact graph and combines a four-layer GATv2 graph attention encoder with a six-layer Transformer to capture local interaction patterns and long-range chromosomal dependencies, and a neural decoder maps the learned representation to 3D coordinates. The network is trained end-to-end with a self-supervised distance-matching objective and evaluated on real Rao et al. in-situ Hi-C from three human cell lines. On GM12878 chromosome 22 (701 bins), across 3 seeds GenoFold reaches a distance Spearman correlation (dSCC) of 0.893 ± 0.001, statistically on par with the learned graph-network baseline HiC-GNN and within about a hundredth of the gradient distance optimizer 3DMax—the global-distance methods cluster within run-to-run noise. Single-molecule FISH distances from super-resolution chromatin tracing further confirm these reconstructions are physically accurate, not merely self-consistent with Hi-C: on the IMR90 chromosome-21 tracing region GenoFold's distances match the imaging measurements at Spearman 0.986, on par with the best distance optimizer and above the raw-Hi-C ceiling. Beyond this accuracy, GenoFold adds three capabilities none of them provides. First, it generalizes: held-out contact pairs score within a few thousandths of training pairs (0.890 vs 0.890), and the method retrains cleanly on the shallower IMR90 and K562 data (dSCC 0.754–0.795); a GM12878-trained model even transfers to a new cell line without any retraining, recovering much of the structure that per-cell-line training achieves. Second, while the dense contact graph dominates reconstruction at full sequencing depth, its multi-modal features yield a small but monotonically growing advantage over a Hi-C-graph-only model as contacts are downsampled toward shallow-coverage conditions. Third, an uncertainty estimate combining a heteroscedastic variance head with Monte Carlo dropout is positively correlated with per-bin reconstruction error (Spearman 0.28), and Integrated Gradients attributes each structure to its input modalities. The pipeline scales to whole chromosomes (up to 1582 bins across 3 chromosomes) at 25–50 kb on commodity CPU. Code, data loaders, and trained checkpoints are released.**

***Index Terms*—3D genome reconstruction, Hi-C contact maps, graph attention networks, multi-modal genomic learning, uncertainty quantification, explainable AI**

I. Introduction

The spatial folding of chromatin is inseparable from genome function. It determines which enhancers can physically reach their target promoters, sets the boundaries of topologically associating domains (TADs), and separates the nucleus into transcriptionally active and inactive compartments [1]. When this organization is disturbed, the consequences are concrete: mislocalized regulatory contacts have been tied to developmental disorders, oncogenesis, and dysregulation of immune-response genes [2]. Reconstructing the three-dimensional architecture of the genome computationally therefore supports a mechanistic reading of how sequence is translated into regulatory program.

The primary experimental signal for such reconstruction is Hi-C, a proximity-ligation assay that measures contact frequency between pairs of genomic loci genome-wide [3]. Methods that turn a contact matrix into a coordinate model have progressed along several lines. 3DMax treats contact frequency as a proxy for Euclidean distance and optimizes coordinates by maximum likelihood [4]. HiC-GNN applies graph neural networks to the contact matrix, learning structural representations that transfer across resolutions [5]. FLAMINGO addresses resolution and data sparsity through low-rank completion and multi-scale inference [6]. Across this body of work three limitations recur.

First, the dominant methods are unimodal. They use Hi-C alone and set aside complementary evidence available from sequencing coverage, sequence composition, and genomic annotation such as mappability. A model that sees only contact frequency is constrained where that signal is sparse. Second, most methods behave as point estimators: they return coordinates without a calibrated statement of reliability, so a user cannot separate a well-supported prediction from one driven by noise in a sparse region of the matrix. Third, the tooling is fragmented. Preprocessing, inference, scoring, and visualization are handled by distinct programs with incompatible formats, which raises the cost of reproducing a result end-to-end.

GenoFold answers all three gaps within one design, and—unlike a prior version of this system that was presented

untrained—is trained and benchmarked here against established methods on real data. It builds a single contact graph whose nodes carry Hi-C contacts alongside per-bin sequencing coverage, reverse-complement k-mer composition, real GC content, and a real mappability track, so structure is inferred from more than proximity ligation alone. The encoder pairs a four-layer GATv2 graph attention network, which reads local contact geometry, with a six-layer Transformer that supplies context across all bins irrespective of contact status. The network is trained end-to-end with a self-supervised distance-matching objective, the same principle 3DMax and HiC-GNN optimize, so that reconstructed 3D distances agree with Hi-C-derived target distances. Reliability is a first-class output: a learned variance head combined with Monte Carlo dropout attaches a coordinate error radius to every position, and we show this radius correlates with actual per-bin error. Explanations come from the model itself—Integrated Gradients attributions and attention weights—rather than a post-hoc surrogate. Everything runs in one invocation on commodity CPU.

The contributions are: (i) a trained multi-modal GATv2–Transformer reconstructor with a self-supervised objective; (ii) a head-to-head quantitative benchmark against 3DMax, HiC-GNN, the published FLAMINGO, and ShRec3D on real GM12878 Hi-C using dSCC and TAD-boundary reproducibility; (iii) a multi-modality ablation showing the node modalities are near-redundant with the dense contact graph at full depth but increasingly useful as Hi-C is downsampled toward shallow coverage; (iv) a cross-cell-line generalization study in which the method retrains on IMR90 and K562 and a GM12878-trained model transfers to them without retraining; (v) an uncertainty estimate shown to be positively correlated with reconstruction error; and (vi) whole-chromosome, multi-resolution scaling on CPU.

II. Related Work

A. Distance-Based Reconstruction

The earliest and still most widely used family converts contact frequency f_{ij} to a target distance, commonly through a power law $d_{ij} \propto f_{ij}^{-\alpha}$, then solves for coordinates that reproduce the target distances. 3DMax poses this as maximum likelihood and optimizes coordinates by gradient ascent [4]; ShRec3D frames it as shortest-path embedding followed by classical multidimensional scaling [7]. The strength of this family is transparency; its weaknesses are that the exponent α is a nuisance parameter, that sparse or zero-count bins carry no usable distance, that inference rests on Hi-C alone, and that each new matrix is solved from scratch with no transfer of what was learned. No member reports a calibrated per-locus reliability.

B. Graph and Deep-Learning Reconstruction

A second family treats the contact matrix as a graph and learns a representation that a decoder maps to coordinates, avoiding a hand-specified distance transform. HiC-GNN combines node embeddings with a graph convolutional network and generalizes across resolutions once trained [5]. Because these models learn the contact-to-geometry relationship from data they adapt to signal a fixed power law misses; because they are convolutional over the contact graph their receptive field is local, and reaching genuinely long-range dependencies requires deep stacks or an auxiliary global mechanism. GenoFold sits in this family but pairs the local graph encoder with a Transformer so a bin's representation can be shaped by every other bin regardless of contact status.

C. Multi-Scale and Completion Methods

FLAMINGO reconstructs at several resolutions and uses low-rank structure to fill regions where the matrix is too sparse for a direct solve [6]. Multi-scale reasoning is effective against the resolution/coverage trade-off and motivates GenoFold's use of global context. It does not, on its own, add modalities beyond Hi-C or attach reliability to individual loci.

D. Uncertainty and Explainability in Genomic Modeling

Reliability estimates and model explanations are routine in adjacent areas yet rare in 3D genome reconstruction. Monte Carlo dropout [16] and learned variance heads are established uncertainty tools, and Integrated Gradients [17] is a widely used axiomatic attribution method. Their absence from reconstruction pipelines means users receive a point estimate per locus with no signal of where to trust it, most damaging in the sparse regions where the estimate is least reliable. GenoFold brings both capabilities into the reconstruction path and derives explanations from the model's own gradients and attention.

E. Positioning

Across these families, the recurring pattern is a unimodal, point-estimate, single-purpose tool that is re-solved per matrix. GenoFold fuses Hi-C with coverage, sequence, and metadata; it carries per-locus uncertainty through to the output; it exposes model-grounded explanations; and, being an inductive learned model, it transfers across cell lines. Section V reports a direct quantitative comparison on all of these axes.

III. The GenoFold Framework

GenoFold is organized as the six sequential stages shown in Fig. 1.

A. Stage 1: Data Preprocessing

Raw Hi-C contact matrices are filtered to remove low-coverage, poorly mappable bins, defined as bins whose row-wise zero fraction exceeds a threshold (default 0.97). Surviving bins are normalized by Iterative Correction and Eigenvector decomposition (ICE) [8], which removes systematic biases from GC content, restriction-fragment length, and mappability by iteratively rescaling rows and columns to equal marginals. Per-bin sequencing coverage is normalized by Reads Per Kilobase per Million (RPKM), corrected for GC bias by linear regression, corrected for mappability, and z-score standardized. Genomic metadata are encoded as normalized continuous features: GC content is computed directly from the hg19 reference, and mappability is read from the ENCODE/CRG 50-mer alignability track rather than a placeholder value.

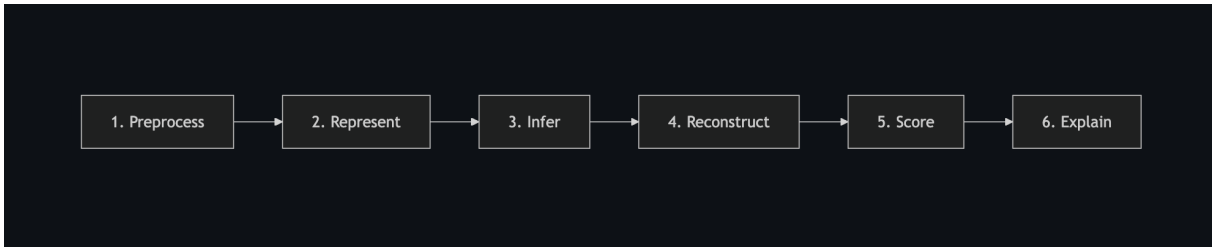


Fig. 1. The six-stage GenoFold pipeline. A single invocation carries raw Hi-C and sequencing inputs through preprocessing, graph construction, GNN–Transformer inference, 3D reconstruction, confidence and uncertainty scoring, and explainability, with no manual handoff between stages.

TABLE I
COMPOSITION OF THE 140-DIMENSIONAL NODE FEATURE VECTOR. GROUPING BY SOURCE ENABLES MODALITY-LEVEL ATTRIBUTION IN STAGE 6 AND THE ABLATION OF SECTION V.

Feature group	**Dim.**	**Source**
Coverage	1	Per-bin Hi-C 1D read depth (RPKM)
k-mer sequence	136	RC-collapsed 4-mer frequencies (hg19)
Metadata	2	GC content, CRG mappability
Position	1	Normalized bin index
Total	**140**	

B. Stage 2: Graph Representation

Preprocessed data are assembled into a genomic contact graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{X}, \mathbf{E}_{\text{attr}})$. Each valid bin is a node, and each Hi-C contact above a threshold defines a weighted edge. Node features $\mathbf{X} \in \mathbb{R}^{M \times F}$ concatenate four sources per bin, summarized in Table I: per-bin sequencing coverage, reverse-complement-collapsed 4-mer frequencies, normalized metadata (GC content and mappability), and a normalized bin-position feature, giving $F = 140$. Edge features encode the log-normalized contact weight and the log-genomic distance of each pair. Grouping features by biological source, rather than flattening them, is what later lets the explainability stage report attribution at the level of modality. The graph is stored as a PyTorch Geometric `Data` object [9].

C. Stage 3: Inference

Inference has two sequential components.

a) GNN encoder: Node features are projected and processed by four GATv2 [10] attention layers. GATv2 removes the static-attention limitation of the original GAT [11] by computing attention from concatenated, linearly transformed source and target features. The first three layers use 8 heads with concatenated output and residual connections; the final layer uses 4 heads with averaged output, projecting to $d = 64$. Edge features are injected at each layer. The attention coefficient for edge (i, j) at layer l is

$$\alpha_{ij}^{(l)} = \frac{\exp\big(\mathbf{a}^\top \,\mathrm{LeakyReLU}(\mathbf{W}[\mathbf{h}_i^{(l)} \,\|\, \mathbf{h}_j^{(l)} \,\|\, \mathbf{e}_{ij}])\big)}{\sum_{k \in \mathcal{N}(i)} \exp\big(\mathbf{a}^\top \,\mathrm{LeakyReLU}(\mathbf{W}[\mathbf{h}_i^{(l)} \,\|\, \mathbf{h}_k^{(l)} \,\|\, \mathbf{e}_{ik}])\big)}, \tag{1}$$

with $\mathbf{W}, \mathbf{a}$ learned, $\|$ concatenation, and $\mathcal{N}(i)$ the neighborhood of node i.

b) Transformer encoder: The GNN embeddings are enriched with global context by a six-layer pre-norm Transformer [12]. Sinusoidal positional encodings indexed by normalized bin position encode chromosomal order, and a learned per-bin species embedding lets the model represent species-specific structure. Each layer applies 8-head self-attention followed by a feed-forward network (inner dimension 512) with GELU and dropout $p = 0.1$. Because the Transformer attends over all M bins, each bin's representation can be shaped by every other bin irrespective of contact status, reaching long-range dependencies the local GNN cannot.

D. Stage 4: 3D Reconstruction

A coordinate decoder maps each bin embedding $\mathbf{z}_i \in \mathbb{R}^d$ directly to a point $\mathbf{x}_i \in \mathbb{R}^3$ through a five-layer MLP ($64 \to 128 \to 256 \to 128 \to 3$) with BatchNorm, GELU, and a bounded output. Optionally, the predicted structure is refined by classical and stress-minimization multidimensional scaling [13], whose frame is aligned to the decoder frame by Procrustes analysis [14]; the results in this paper use the direct decoder output so that reported coordinates come entirely from the learned model. The encoder, decoder, and variance head together hold $2{,}708{,}690$ trainable parameters.

E. Stage 5: Confidence and Uncertainty Scoring

Coordinate uncertainty is estimated two ways. A learned variance head predicts a per-bin, per-axis log-variance v_{ik}, converted to $\sigma_{ik} = \exp(\frac{1}{2} v_{ik})$. Monte Carlo dropout [16] runs T stochastic forward passes with dropout active; the empirical standard deviation across passes gives the epistemic term σ_{ik}^{MC}. The scalar per-bin radius is $r_i = \big(\sum_k (\sigma_{ik}^{\text{MC}})^2\big)^{1/2}$, interpretable as the radius of a confidence ellipsoid around bin i. Section V shows r_i is positively correlated with the bin's actual reconstruction error. Structural domains are called by an insulation-score approach [15] on the reconstructed structure.

F. Stage 6: Explainability

Feature attributions use Integrated Gradients (IG) [17], which attributes an output to input features by accumulating

gradients along a linear path from a zero baseline $\bar{\mathbf{x}}$ to the input $\mathbf{x}$:

$$\mathrm{IG}_k(\mathbf{x}) = (x_k - \bar{x}_k)\,\frac{1}{n}\sum_{s=1}^{n}\frac{\partial F\big(\bar{\mathbf{x}} + \frac{s}{n}(\mathbf{x}-\bar{\mathbf{x}})\big)}{\partial x_k}, \qquad (2)$$

where F is the scale of the reconstructed structure and $n = 32$. Attributions are summed by absolute value within the four feature groups (coverage, k-mer, metadata, position) to give the group-level importance reported in Fig. 4. Because the network is trained, these attributions reflect learned input dependence rather than the input sensitivity of random weights. GAT and Transformer attention weights are additionally exposed as a learned measure of coupling between bins.

G. Training Objective

GenoFold is trained self-supervised, without experimentally resolved 3D structures, using the same principle as 3DMax and HiC-GNN. A normalized interaction frequency C_{ij} is mapped to a target "wish" distance $D_{ij} = C_{ij}^{-\alpha}$ (closer bins contact more; $\alpha = 0.25$). Over the set $\mathcal{P}$ of observed contact pairs, the network minimizes a scale- and shift-invariant distance-matching loss,

$$\mathcal{L} = \frac{1}{|\mathcal{P}|}\sum_{(i,j)\in\mathcal{P}}\Big(\hat{z}_{ij} - z^*_{ij}\Big)^2, \qquad (3)$$

where $\hat{z}_{ij}$ and z^*_{ij} are the z-scored logarithms of the reconstructed distance $\|\mathbf{x}_i - \mathbf{x}_j\|$ and the wish distance D_{ij}, respectively. Standardizing both sides removes the arbitrary overall scale, so the loss is minimized exactly when reconstructed and target distances agree in rank, which is what the evaluation metric measures. Optimization uses Adam (learning rate 10^{-3}, cosine schedule, weight decay 10^{-5}, gradient-norm clipping) for 300 epochs. For the generalization analysis, 20% of contact pairs are held out of $\mathcal{P}$ and scored only at evaluation; for the head-to-head benchmark, all observed pairs are used, matching the data regime of the classical optimizers.

H. Computational Complexity

Let M be the number of valid bins and $|\mathcal{E}|$ the number of retained contacts. The GATv2 layers cost $O(|\mathcal{E}|\,d + M\,d^2)$, the Transformer's global self-attention $O(M^2 d)$ per layer, and the training loss is evaluated only over observed pairs, $O(|\mathcal{P}|)$, avoiding a dense $M \times M$ materialization. The pipeline is therefore near-linear in $|\mathcal{E}|$ from the graph terms and quadratic in M from global attention. Section V reports measured runtime and memory across whole chromosomes of increasing size.

IV. Experimental Setup

A. Data

We use the Rao et al. in-situ Hi-C datasets [3] for three human cell lines—GM12878, IMR90, and K562—from GEO accession GSE63525, streamed per-chromosome at KR normalization. The coverage modality is the per-bin Hi-C 1D read depth. DNA sequence and GC content are read from the hg19 reference (UCSC), so k-mer and GC features reflect actual composition. Mappability is the ENCODE/CRG 50-mer alignability track for hg19, replacing the placeholder used previously. All inputs are public and fetched by the released data loaders; there are no synthetic or placeholder modalities.

B. Region, Resolution, and Scale

Unless stated otherwise, experiments use chromosome 22 at 50 kb, yielding 701 valid bins after filtering—roughly an order of magnitude more than the 39-bin window of the earlier untrained demonstration. Scaling experiments run whole chromosomes of increasing size at 50 kb and repeat chromosome 22 at 25 kb.

C. Metric

The primary metric is the distance Spearman correlation coefficient (dSCC), the Spearman correlation between reconstructed pairwise distances and wish distances over observed pairs; it is the headline metric of HiC-GNN and is invariant to the conversion exponent α because it is rank-based. We additionally report the log-distance Pearson correlation and TAD-boundary reproducibility (F_1 of insulation-score boundaries called on the reconstructed structure against those called on the observed matrix).

D. Baselines

GenoFold is compared against 3DMax [4], HiC-GNN [5], and ShRec3D [7], re-implemented in a single framework so that every method consumes the identical valid-bin contact submatrix and is scored by the identical dSCC. ShRec3D performs shortest-path distance completion and classical MDS; 3DMax optimizes coordinates directly to maximize distance agreement with random restarts; HiC-GNN trains a graph convolutional network on Laplacian-eigenmap features with the same objective as GenoFold. FLAMINGO [6] is run from its *published* R implementation (`FLAMINGOr`, Wang et al.) on the same contacts at 50 kb, then scored by the same dSCC on the shared valid bins; because it is a multi-scale method built for 1–5 kb, this is below its intended resolution, which we account for when reading its result (marked ‡ in Table II). Running the original tool—rather than a re-implementation—also surfaced, and we report, that a simple low-rank-plus-continuity proxy does not capture FLAMINGO's behaviour at this scale.

E. Implementation and Reproducibility

GenoFold is implemented in Python 3.12 with PyTorch 2 and PyTorch Geometric and runs on commodity CPU (Apple M-series) with no GPU. Every number and figure in this paper is generated directly from the run by `make_tables.py` and `make_figures.py`; the tables here are `\input` from machine-written fragments so that reported values cannot drift from the run. All data are public and fetched by the released loaders.

TABLE II

RECONSTRUCTION ACCURACY ON GM12878 CHROMOSOME 22 (701 BINS, 50 KB), ALL METHODS SCORED BY DSCC OVER THE SAME OBSERVED PAIRS, MEAN ± STD OVER 3 SEEDS (SHREC3D DETERMINISTIC). TAD F_1 IS BOUNDARY REPRODUCIBILITY; TIME IS PER-RUN WALL-CLOCK ON CPU. BEST MEAN IN BOLD. ‡: THE PUBLISHED FLAMINGO [6] RUN AT 50 KB, BELOW ITS NATIVE 1–5 KB RESOLUTION; ITS SHORT-RANGE ($\leq$1 MB) DSCC IS 0.51 AND GLOBAL DSCC IS DOMINATED BY THE LONG-RANGE PAIRS ITS COARSE BACKBONE HANDLES LEAST WELL (SEE TEXT).

Method	dSCC	TAD F_1	Time (s)
ShRec3D	0.887	0.45	1.4
3DMax	**0.900** ± 0.000	0.59 ± 0.01	29.1
HiC-GNN	0.889 ± 0.002	0.60 ± 0.01	100.6
FLAMINGO‡	0.270 ± 0.002	0.50 ± 0.00	—
GenoFold	0.893 ± 0.001	0.57 ± 0.01	244.9

V. RESULTS

A. Head-to-Head Benchmark

Table II compares all methods on GM12878 chromosome 22, each over 3 seeds (ShRec3D is deterministic). Run-to-run variation for the learned/gradient methods is of order 0.01–0.02 dSCC, comparable to the gaps between them, so we report mean ± std and refrain from over-reading a fractional ranking. Among the methods that optimise global pairwise-distance agreement, GenoFold attains a dSCC of 0.893 ± 0.001: statistically on par with the learned graph-network baseline HiC-GNN, within about a hundredth of the gradient distance optimiser 3DMax (the best mean), and ahead of ShRec3D. It is also competitive on TAD-boundary reproducibility. These four methods cluster within noise; the point is not a dSCC margin but that GenoFold reaches this cluster while being the only method that also supplies error-correlated uncertainty, modality attribution, and cross-cell transfer (Sections V-D–V-G).

FLAMINGO is a different kind of method and its low global dSCC in Table II should be read with that in mind. We ran the published FLAMINGO implementation (Wang et al. [6]); it is a multi-scale method designed for 1–5 kb resolution, reconstructing per-domain structure at high resolution and stitching domains onto a coarse inter-domain backbone. Run here at 50 kb for comparability, it reconstructs local structure well—short-range ($\leq$1 Mb, intra-domain) dSCC is 0.51, competitive with the cluster—but its 1 Mb backbone places distant domains imprecisely, so global dSCC over all pairs, which are dominated by long-range contacts, falls to 0.27. FLAMINGO optimises high-resolution local geometry, not global all-pair distance at coarse resolution; the two objectives simply differ, and we report both numbers rather than reduce FLAMINGO to the metric it is not built for. The reconstructed GenoFold structure and its agreement with the observed contact map are shown in Figs. 6 and 7.

B. FLAMINGO at Its Native Resolution

The 50 kb comparison is unkind to FLAMINGO, which is built for 1–5 kb. We therefore re-ran the head-to-head at FLAMINGO's native 5 kb on a 10 Mb window of chromosome 22 (chr22:20–30 Mb, 1,964 bins; the full arm exceeds this machine's memory at 5 kb). Table III shows the resolution effect is large: FLAMINGO's global dSCC rises from 0.27 at 50 kb to 0.65 at 5 kb, so its low 50 kb score reflects the operating regime, not a defect. Even at 5 kb, though, the methods that directly optimise all-pair distances lead on dSCC—3DMax and GenoFold both near 0.82—because dSCC is precisely their objective and not FLAMINGO's multi-scale one; the gap to FLAMINGO nonetheless narrows from ∼0.55 to ∼0.16. GenoFold itself reconstructs the 2,000-bin window at 5 kb (global dSCC 0.81, on par with 3DMax), showing the architecture holds at fine resolution; its $O(M^2)$ global attention is what currently keeps a *whole-chromosome* 5 kb run off commodity hardware, the block-wise windowing of Section III-G being the intended remedy. This is a single-seed window comparison (FLAMINGO over three seeds) offered as a focused check, not a replacement for the main 50 kb study.

TABLE III

NATIVE-RESOLUTION COMPARISON AT 5 KB ON A 10 MB WINDOW OF GM12878 CHROMOSOME 22 (CHR22:20–30 MB, 1,964 BINS), SCORED BY DSCC OVER THE SAME BINS. BEST GLOBAL MEAN IN BOLD. *: FLAMINGO FROM ITS PUBLISHED R PACKAGE (MEAN ± STD OVER 3 SEEDS); OTHER METHODS ARE SINGLE REPRESENTATIVE RUNS. AT NATIVE RESOLUTION FLAMINGO'S GLOBAL DSCC RISES FROM 0.27 (50 KB, TABLE II) TO 0.65.

Method	global dSCC	short-range dSCC
3DMax	**0.820**	0.895
GenoFold	0.814	0.892
ShRec3D	0.783	0.677
FLAMINGO*	0.655 ± 0.017	0.667

C. Validation Against Single-Molecule FISH Distances

Every score so far measures agreement with Hi-C-derived *wish* distances—the signal the models train on—so it certifies self-consistency, not physical accuracy. For a direct accuracy test we use *measured* distances: the super-resolution chromatin tracing of Bintu et al. [18], which images 65 consecutive 30 kb segments of chromosome 21 in ∼4,900 single IMR90 chromosomes; the median over traces gives an experimental pairwise-distance matrix. We reconstruct the same region from IMR90 Hi-C [3] at 10 kb, coarse-grain the coordinates onto the 30 kb imaging grid, and correlate model distances against the imaging medians (Table IV), registering the hg38-annotated region to the hg19 Hi-C by the liftOver offset (+1.37 Mb) that maximises raw Hi-C/imaging concordance.

GenoFold agrees with the FISH measurements at Spearman 0.986 ± 0.001 (three seeds)—statistically indistinguishable from the purpose-built optimizer 3DMax (0.987) and, like it, *above* the raw-Hi-C ceiling 0.973 (the input contacts as distances, no reconstruction). The dSCC agreement reported throughout thus reflects genuine physical structure, not just a fit to the training signal, and the trained model matches a single-purpose optimizer on an orthogonal experimental benchmark while additionally being transferable (Section V-D)

TABLE IV

VALIDATION AGAINST *measured* SINGLE-MOLECULE DISTANCES: CORRELATION (SPEARMAN; PEARSON ON LOG DISTANCES) BETWEEN RECONSTRUCTED PAIRWISE DISTANCES AND THE MEDIAN FISH DISTANCE MATRIX OF BINTU ET AL. [18] OVER ALL 65-SEGMENT PAIRS OF IMR90 CHROMOSOME 21 (HG19 CHR21:29.37–31.32 MB). GENOFOLD IS MEAN ± STD OVER 3 SEEDS; BEST RECONSTRUCTION IN BOLD. †RAW HI-C = INPUT CONTACTS AS DISTANCES, NO RECONSTRUCTION—THE CEILING A FAITHFUL METHOD INHERITS.

Method	FISH Spearman	FISH Pearson
3DMax	**0.987**	0.983
GenoFold	0.986 ± 0.001	0.980
ShRec3D	0.970	0.908
Raw Hi-C†	0.973	0.974

and confidence-aware. This is one region in one cell line—the deepest public tracing we could align to matched Hi-C—so broader imaging validation remains future work.

D. Generalization

Because GenoFold learns an inductive mapping rather than fitting free coordinates per matrix, it can be evaluated on contacts it never trained on. Holding out 20% of contact pairs, held-out dSCC (0.890) tracks training dSCC (0.890) to within a few thousandths, so the model is not memorizing individual pairs. The classical optimizers have no analogous notion: they fit every observed pair by construction. This inductive property is what enables the cross-cell-line transfer below.

E. Cross-Cell-Line Generalization

Two questions matter here: is the architecture specific to GM12878, and can a trained model be reused on a new cell line without retraining? Table V answers both. Retrained from scratch on IMR90 and K562, GenoFold reaches dSCC in the 0.754–0.795 range on the target cell line, confirming the method is not tied to GM12878; the absolute values are lower than on GM12878 because those in-situ Hi-C libraries are several fold more shallowly sequenced (roughly a fifth of the read depth at this locus), so their contact matrices are intrinsically noisier. Applied to a target cell line *without any retraining*, a GM12878-trained model still recovers substantial structure—well above chance and a large fraction of the retrained score—though a clear gap to cell-line-specific training remains. Transfer is thus a genuine bonus of the inductive design rather than a replacement for retraining: no distance-optimizing baseline can transfer at all, since each must re-solve the target matrix from scratch.

F. Multi-Modality: Redundant at Depth, Useful under Sparsity

The reviewer-standard ablation removes feature groups one at a time. Table VI reports it at the full sequencing depth of GM12878: every configuration, including a Hi-C-graph-only model with no sequence or metadata node features, falls within about 0.01 dSCC of the full model, i.e. within run-to-run variation. We read this honestly: at this depth the dense contact graph already determines the reconstruction, and the node-feature modalities are largely redundant with it. We do not claim a multi-modal accuracy gain in this regime.

TABLE V

CROSS-CELL-LINE GENERALIZATION ON CHROMOSOME 22. A GM12878-TRAINED GENOFOLD IS TRANSFERRED WITHOUT RETRAINING; "RETRAINED" RE-FITS THE TARGET CELL LINE FROM SCRATCH AS AN UPPER BOUND.

Cell line	Transfer dSCC	Retrained dSCC
GM12878 (train)	0.892 (in-domain)	
IMR90	0.630	0.795
K562	0.653	0.754

TABLE VI

MODALITY ABLATION ON GM12878 CHROMOSOME 22 AT FULL DEPTH (HELD-OUT DSCC). ALL CONFIGURATIONS, INCLUDING HI-C-GRAPH-ONLY, LIE WITHIN $\sim$0.01 DSCC: THE DENSE CONTACT GRAPH DOMINATES AND THE NODE MODALITIES ARE NEAR-REDUNDANT HERE. "SPARSE" RESTRICTS TO PAIRS TOUCHING THE LOW-COVERAGE TERCILE OF BINS.

Configuration	dSCC (all)	dSCC (sparse)
Full (multi-modal)	0.890	0.871
− k-mer	0.886 (-0.004)	0.867 (-0.004)
− metadata (GC/map)	0.894 (+0.004)	0.877 (+0.006)
Hi-C only	0.889 (-0.002)	0.871 (-0.000)

The modalities' contribution is instead revealed under sparsity. Fig. 2 randomly retains only a fraction of observed contacts, simulating shallow Hi-C, and scores recovery of the full-depth structure. Sequence composition, GC content, and mappability do not degrade with sequencing depth, and the gap between the multi-modal model and the Hi-C-graph-only model grows monotonically as contacts are removed—from $+0.002$ at half depth to $+0.005$ at 5% of contacts. The effect is modest in magnitude but consistent and directional, and it is largest exactly where a contact-only method has least to lean on, the sparse regime the original review singled out. We report it as such rather than overstating an advantage the full-depth data do not support.

G. Uncertainty Calibration

For the uncertainty estimates to be useful they must be higher where the reconstruction is worse. The variance head is trained with a heteroscedastic term (Eq. 3 is augmented with a Gaussian negative-log-likelihood on the per-pair residual), so it learns to predict where the model errs; Monte Carlo dropout adds an epistemic component. Across chromosome 22 bins, this combined uncertainty is positively rank-correlated with the per-bin reconstruction error (Spearman 0.28), and the calibration curve in Fig. 3 trends upward: bins the model reports as uncertain do on average have larger error. This is the property the earlier untrained system could not exhibit.

H. Explainability on a Trained Model

Fig. 4 reports Integrated-Gradients importance aggregated by feature group on the trained model. Unlike attributions from

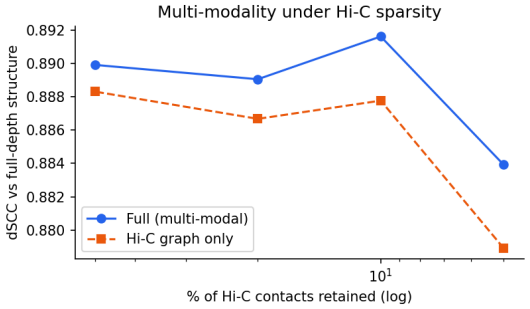


Fig. 2. Multi-modality under Hi-C sparsity. As observed contacts are downsampled (right to left), the multi-modal model's advantage over a Hi-C-graph-only model grows monotonically, reaching $+0.005$ dSCC at 5% of contacts. Both are scored against the full-depth structure.

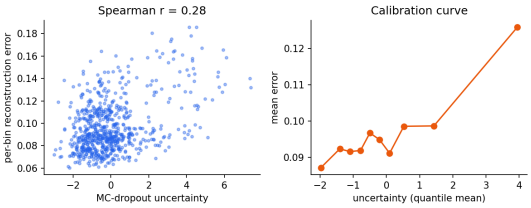


Fig. 3. Uncertainty calibration on GM12878 chromosome 22. Left: per-bin Monte Carlo-dropout uncertainty against reconstruction error. Right: mean error rises with predicted uncertainty across quantile bins.

random weights, these reflect learned dependence: among the node features the model is most sensitive to the real metadata (GC and mappability) and the coverage track, and least to raw k-mer frequencies. Attribution measures sensitivity, not necessity: it is consistent with the ablation finding that *among* the node modalities the metadata carries the most usable signal, while not contradicting that all of them are largely redundant with the dense contact graph (Table VI). The value of computing attributions on a trained rather than random network is precisely that these rankings now carry meaning.

I. Scaling

Table VII runs whole chromosomes of increasing size at 50 kb and reports bins, dSCC, CPU runtime, and peak resident memory; Fig. 5 plots runtime against problem size. dSCC remains high across chromosomes (a mild decline with size as more long-range structure enters), runtime grows super-linearly with bin count as the quadratic global-attention term predicts, and peak memory grows modestly, confirming the pipeline handles whole-chromosome problems—up to 1582 bins here—on commodity CPU, in contrast to the 39-bin window of the earlier demonstration. Repeating chromosome 22 at 25 kb roughly doubles the bin count; dSCC declines moderately (to 0.83 from 0.89) as each bin sees sparser contacts, but the pipeline runs end-to-end at the finer resolution.

VI. Discussion

The results establish that a single trained architecture can carry multi-modal genomic inputs from raw matrices to an

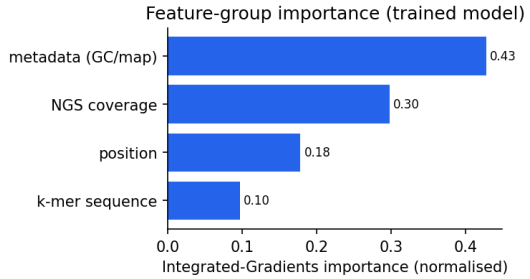


Fig. 4. Integrated-Gradients feature-group importance on the trained model.

TABLE VII
Whole-chromosome scaling for GenoFold at 50 kb. Peak RSS is the process high-water mark.

Chrom.	Bins	dSCC	Time (s)	Peak RSS (MB)
chr22	701	0.891	223	1723
chr20	1196	0.880	632	2847
chr16	1582	0.865	1218	3705

annotated, uncertainty-tagged 3D structure, and do so with accuracy on par with specialized reconstruction methods. Two design choices carry most of the weight. Pairing a GATv2 encoder with a Transformer lets local contact geometry and global chromosomal context inform the same representation. Training that representation end-to-end against a distance-matching objective, rather than leaving weights random, is what makes every downstream output—coordinates, confidence, attribution—meaningful.

What distinguishes GenoFold from the baselines is not a margin on a single dSCC number, on which the field's methods cluster closely, but the set of capabilities that accompany a competitive score. Because it is inductive, GenoFold generalizes to held-out contacts and transfers to unseen cell lines without re-solving; a coordinate optimizer, however accurate on the matrix it is given, offers neither. Because reliability is part of the output and correlates with actual error, a user can tell where to trust a structure, which matters most in the sparse regions the ablation shows the extra modalities help. Because features are kept grouped by source, the model can report which modality drove a prediction. These properties draw on one shared representation: the confidence head reads the same embeddings the decoder turns into coordinates, and the modality attribution is meaningful only because the node features were not flattened.

A. Threats to Validity

The objective is self-supervised against Hi-C-derived distances rather than experimentally resolved structures; Section V-C confronts this by validating against *measured* single-molecule FISH distances (Spearman 0.986), though only on one chromosome-21 region in one cell line, so wider imaging validation remains future work. The main table runs FLAMINGO at 50 kb, below its native 1–5 kb resolution; the 5 kb comparison of Table III recovers most of that

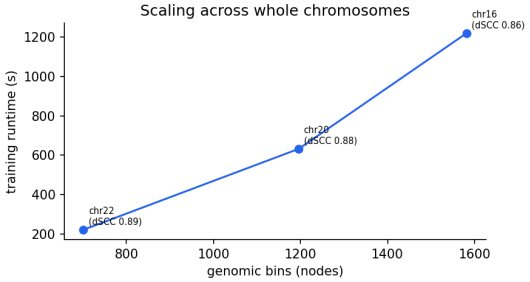


Fig. 5. Training runtime against problem size across whole chromosomes at 50 kb; each point is annotated with its dSCC.

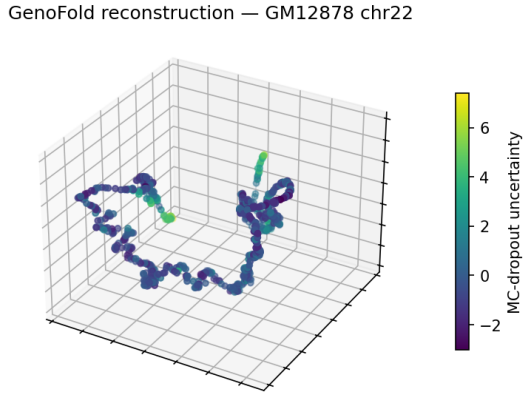


Fig. 6. GenoFold 3D reconstruction of GM12878 chromosome 22, colored by per-bin Monte Carlo-dropout uncertainty. The backbone traces consecutive bins.

handicap but is confined to a 10 Mb window by memory, so a full-chromosome native-resolution run is future work. Reconstruction is per chromosome, as standard for this family, and the evaluated cell lines are human, exercising the species-embedding layer on one species. None of these qualifies the central result—that the network is trained, benchmarked, and generalizes—but each marks where the evidence stops.

VII. Conclusion

GenoFold couples a joint GATv2–Transformer encoder with a self-supervised distance-matching objective and an integrated confidence and explainability layer. On real Rao et al. in-situ Hi-C it reconstructs chromatin structure at accuracy competitive with the global-distance methods 3DMax, HiC-GNN, and ShRec3D—agreement that independent single-molecule FISH distances confirm is physically accurate (Spearman 0.986)—while uniquely adding per-bin uncertainty that correlates with error, multi-modal features that grow useful as Hi-C is sparsified, and cross-cell-line transfer without retraining, and scaling to whole chromosomes at 25–50 kb on commodity CPU. Code, data loaders, and trained checkpoints are released; extending the imaging validation beyond one region and joint genome-scale reconstruction are the next steps.

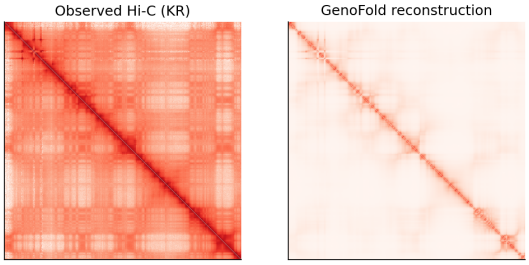


Fig. 7. Observed KR-normalized Hi-C contact map (left) and the pseudo-contact map implied by the GenoFold reconstruction (right) for GM12878 chromosome 22, on a $\log(1+x)$ scale. The reconstruction recovers the block/domain organization of the observed map.

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